Statistical Validation of the Formulon Physics Legendre Encoder: Shot-Noise Scaling, Decoder Efficiency and Deterministic Accuracy on a PennyLane Simulator

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ABSTRACT

Formulon Physics (v1.0.0) provides a Legendre-polynomial feature encoder whose features are read out as Pauli-Z expectation values on a PennyLane `default.qubit` simulator, together with a decoder that reconstructs the input x from the measured features. This preprint reports an independent, reproducible benchmark of `formulon.quantum_computing.legendre_encoder` executed on Google Colab on 29 October 2026. We (i) compare the deterministic features with SciPy's `eval_legendre` over 1,312 $(x, \text{degree}, \text{order})$ triples and (ii) run 4,100 finite-shot calls of `measure_and_decode` across 41 inputs in $[-1,1]$, five degrees $(2,4,6,8,12)$, four shot budgets (100–5000) and five seeded repetitions, yielding 26,240 feature-level records. The deterministic features agree with SciPy to a maximum absolute difference of 1.03×10^{-15} (float64 round-off). The finite-shot feature error follows binomial shot noise: the observed RMSE equals the theoretical value $\sqrt{\langle 1-P_n^2\rangle/N}$ to within $-5\,\%$ to $+4\,\%$, log–log slopes against shots lie between -0.49 and -0.52, standardised errors have mean 0.008 and standard deviation 0.995, and no systematic bias is detected. Decoded-input error scales as $\sigma_d/\sqrt{N}$ with σ_d falling from 0.804 (degree 2) to 0.155 (degree 12), and the decoder attains the Cramér–Rao bound within about $\pm8\,\%$ in all 20 conditions. The study is an ideal-simulator verification; it makes no claim of quantum advantage and does not characterise hardware noise.

KEYWORDS Formulon Physics; Legendre polynomials; quantum feature encoding; shot noise; Cramér–Rao bound; PennyLane; Google Colab; reproducibility.

I. INTRODUCTION

Encoding a classical scalar into a quantum register is the first step of most variational and hybrid quantum–classical learning pipelines. Polynomial feature maps are attractive because they give the model a controlled, interpretable basis; Legendre polynomials [1] are orthogonal on $[-1,1]$ and numerically well conditioned, which makes them a natural choice for bounded inputs. Formulon Physics [9] is an open Python library of physics formulas and quantum-computing utilities. Its Legendre encoder produces the features $P_0(x),\ldots,P_{d-1}(x)$ for a requested degree d and exposes them as single-qubit Z expectation values that must be estimated from a finite number of measurement shots.

A user of such an encoder needs answers to three practical questions. First, is the deterministic mathematics correct, i.e. does the library return the same Legendre values as an independent reference? Second, how does the statistical error of the measured features depend on the shot budget N, and is it consistent with the theoretical binomial noise floor? Third, how accurately can the input x be recovered by the supplied decoder, and is that decoder statistically efficient or does it leave information unused? Answering these on the library's own public API, rather than on a re-implemented circuit, is the purpose of this report.

Contributions.

- A reproducible Colab benchmark that calls the published Formulon API directly (no substitute circuit) and releases all raw records, summaries, figures and metadata.
- A deterministic cross-check against SciPy [8] over 1,312 values, with maximum difference 1.03×10^{-15}.
- A closed-form shot-noise prediction that reproduces the observed feature RMSE in all 20 (degree, shots) conditions to about $\pm5\,\%$, plus normality and bias diagnostics.
- A Cramér–Rao efficiency analysis of the decoder and a shot-budget table translating a target accuracy into a required number of shots.
- An explicit list of limitations, including a seed-reuse detail in the benchmark script.

Scope. The benchmark uses an ideal (noise-free) simulator with finite sampling and therefore characterises *statistical* error only. Results should not be extrapolated to hardware without a separate noise study, and nothing here tests or implies quantum advantage.

The paper is organised as follows. Section II summarises the mathematical background and derives the shot-noise prediction. Section III describes the experimental design and metrics. Section IV presents the results. Sections V and VI discuss implications and limitations, and Section VII documents reproducibility and the AI-assistance disclosure.

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II. BACKGROUND AND THEORETICAL PREDICTIONS

A. Legendre features. The Legendre polynomials $P_n(x)$ satisfy $|P_n(x)| \le 1$ on $[-1,1]$, with $P_n(1) = 1$ and $P_n(-1) = (-1)^n$, and obey Bonnet's recurrence

$$(n+1)P_{n+1}(x) = (2n+1)\,x\,P_n(x) - n\,P_{n-1}(x), \qquad (1)$$

with $P_0 = 1$ and $P_1 = x$. A feature vector of requested degree d is $\mathbf{f}(x) = [P_0(x), \ldots, P_{d-1}(x)]$. The boundedness $|P_n| \le 1$ allows each feature to be represented as the expectation value $\langle Z \rangle \in [-1,1]$ of a qubit observable.

B. Shot-noise model. If a feature is read out as the mean of N independent ± 1 outcomes with expectation $p = P_n(x)$, the estimator $\hat{p}$ is an unbiased binomial mean with

$$\mathrm{Var}(\hat{p}) = \frac{1-p^2}{N}. \qquad (2)$$

Two consequences follow. (i) The noise vanishes when $|p| = 1$, so $P_0 \equiv 1$ is always exact and the endpoints $x = \pm 1$ are exact for every n. (ii) Averaged over a set of inputs, the root-mean-square feature error of a degree-d encoder is

$$\mathrm{RMSE}_{\mathrm{feat}}(d,N) = \sqrt{\frac{\langle 1 - P_n(x)^2 \rangle_{x,n}}{N}} \propto N^{-1/2}. \qquad (3)$$

The average $\langle 1 - P_n^2 \rangle$ rises with d because the exactly known P_0 becomes a smaller fraction of the features and because the mean of P_n^2 over a uniform grid tends to $1/(2n+1)$. We therefore expect the feature RMSE to *increase* with degree at fixed N and to approach $1/\sqrt{N}$ from below.

C. Decoding and the Cramér–Rao bound. The decoder maps the measured feature vector to an estimate $\hat{x}$. For independent, approximately Gaussian feature errors with variances (2), the Fisher information about x is

$$I(x) = N \sum_{n=1}^{d-1} \frac{[P_n'(x)]^2}{1 - P_n(x)^2}, \qquad (4)$$

and any unbiased estimator satisfies $\mathrm{Var}(\hat{x}) \ge 1/I(x)$ [3, 7]. Because $I(x)$ grows with every additional feature, the bound predicts that higher degrees *decrease* the input error even though they *increase* the per-feature error, and that both scale as $N^{-1/2}$. For $d = 2$ the only informative feature is $P_1(x) = x$, so the decoded error equals the P_1 error and the feature RMSE (which averages in the noiseless P_0) is exactly the decoded RMSE divided by $\sqrt{2}$.

We evaluate the bound on interior points ($|x| < 0.95$) because the Fisher information is singular at the endpoints, where all variances vanish. The comparison treats Formulon's decoder as a black box; no assumption is made about its internal algorithm.

III. METHODS

A. Software and environment. All numerical work was executed on Google Colab on 29 October 2026 using the script `formulon_legendre_benchmark_colab.py`. The recorded environment (Table 7) comprises Formulon Physics 1.0.0, PennyLane [2] 0.45.1 (`default.qubit`), SciPy 1.16.3, NumPy [4] 2.1.3, pandas [6] 2.2.3 and Python 3.13 on Linux (x86_64). The script imports `legendre_features` and `measure_and_decode` from `formulon.quantum_computing.legendre_encoder` and does not replace them with a separately written circuit; a Qiskit cross-check is left for future work.

B. Experimental design. The design is fully factorial (Table 1). Seeds were generated deterministically as seed $= 20261028 + 10^5\,i_x + 10^3\,d + 10N + r$, with i_x the input index and r the repetition, so the study can be regenerated bit-for-bit given the same library versions. The encoder was called with `xmin=-1` and `xmax=1`.

C. Metrics. For every feature record the signed error is $\varepsilon = \hat{f}_n - P_n(x)$, with P_n from SciPy. For each (d,N) cell we report the mean absolute error (MAE) $\langle|\varepsilon|\rangle$, the root-mean-square error (RMSE) $\sqrt{\langle\varepsilon^2\rangle}$, the maximum absolute error, the bias $\langle\varepsilon\rangle$, and the same MAE/RMSE/maximum for the decoded input $|\hat{x} - x|$. Standardised errors $z = \varepsilon/\sqrt{(1-P_n^2)/N}$ test the binomial model; the 5 020 records with exactly zero predicted variance (P_0 and endpoint values), whose observed error is exactly zero, are excluded.

D. Statistical procedures. Log–log slopes are least-squares fits of RMSE against N for each degree. Bias significance uses $z = \mathrm{mean}/\mathrm{SE}$ per cell with the standard error from the within-cell standard deviation; because 20 cells are tested, a Bonferroni factor of 20 is applied when interpreting the largest statistic. The Cramér–Rao comparison uses the per-x bound averaged over $|x| < 0.95$ and the empirical decoded MSE over the same points. All derived quantities were computed from the released CSV files with NumPy, pandas and SciPy; figures were produced with Matplotlib [5].

IV. RESULTS

A. Deterministic agreement with SciPy. Before any sampling, Formulon's deterministic `legendre_features` output was compared with `scipy.special.eval_legendre` for every x, degree and order (1,312 comparisons). The maximum absolute difference is 1.03×10^{-15}, the mean is 6.83×10^{-17}, and 702 of 1,312 values (53.5 %) are bit-identical. All nonzero differences lie at the 10^{-16}–10^{-15} level, a few units in the last

TABLE 1. Factorial design. Each call yields d feature records and one decoded x, so the raw table has 26 240 rows and no missing values.

Factor	Levels	Count
Input x	41 points in $[-1,1]$, step 0.05	41
Degree d	2, 4, 6, 8, 12	5
Shots N	100, 500, 1000, 5000	4
Repetitions	independent seeds per cell	5
Total calls	$41 \times 5 \times 4 \times 5$	4 100
Feature records	$41 \times 32 \times 4 \times 5$	26 240

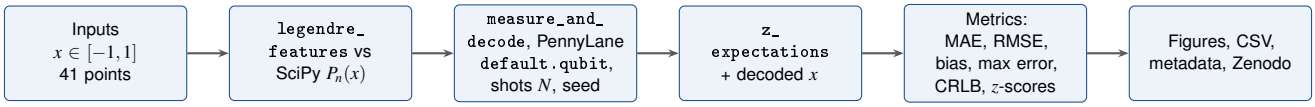


FIG. 1. Benchmark workflow. Deterministic features are first validated against SciPy; finite-shot calls then produce measured features and a decoded input, from which all metrics and figures are derived.

TABLE 2. Maximum deterministic difference between Formulon and SciPy by requested degree.

Degree	max $\lvert\Delta\rvert$	Values compared
2	0 (exact)	82
4	6.66×10^{-16}	164
6	9.99×10^{-16}	246
8	1.03×10^{-15}	328
12	1.03×10^{-15}	492

place of float64, as expected when mathematically equivalent recurrences are evaluated in different orders. The maximum difference grows only marginally with degree (Table 2), so there is no sign of recurrence instability up to $d = 12$.

B. Finite-shot feature reconstruction. Figure 3 shows the measured mean of each Legendre feature over the five repetitions at $d = 6$ and $N = 5000$, overlaid on the SciPy curve. The measured means lie on the reference curves across the full range of x; P_0 is exactly 1 with no scatter, as predicted in Sec. II B, and no deviation is visible at the plotted scale (the expected standard error of the mean at $N = 5000$ over five repetitions is below 0.01).

C. Accuracy versus shot count. Figures 4 and 5 show that both error measures fall monotonically with shots for every degree. Feature RMSE rises with degree (0.0588 at $d = 2$ to 0.0875 at $d = 12$ for $N = 100$) whereas decoded-input RMSE falls (0.0831 to 0.0149). This opposite ordering is what Secs. II B and II C predict: each feature becomes slightly noisier on average, but the larger feature set carries far more information about x. All 20 conditions are listed in Table 3.

D. Agreement with the binomial shot-noise model. For each cell we computed the theoretical RMSE of Eq. (3) from the true feature values and compared it with the observed RMSE. The ratio observed/theory has mean 0.998 and standard deviation 0.024 across the 20 cells, with range 0.949–1.043 (Fig. 6, Table 4) and no trend with degree or shots. Fitted log–log slopes of feature RMSE against N are -0.518, -0.497, -0.487, -0.500 and -0.496 for $d = 2, 4, 6, 8, 12$, all within 0.04 of the theoretical -0.5; for decoded x the slopes lie between -0.47 and -0.52.

E. Error distribution and bias. Across the 21 220 records with nonzero predicted variance, the standardised error z has mean 0.008, standard deviation 0.995, skewness 0.006 and excess kurtosis 0.08, all consistent with a standard normal distribution (Fig. 7). The histogram and Q–Q plot show no heavy tails. Together with the ratio test, this supports the interpretation that the dominant error source is independent binomial sampling noise with the predicted variance.

Bias estimates (Fig. 8) are small: the mean signed error in every cell is below 0.0031 in magnitude, against typical RMSEs of 0.008–0.088. Nineteen of the 20 cells have $|z| < 1.5$. The largest statistic is $z = 3.28$ for $d = 8$, $N = 500$ (mean $+0.0030$, SE 0.0009); with 20 tests this gives a Bonferroni-adjusted $p \approx 0.02$, and the neighbouring cells at the same degree ($N = 100, 1000, 5000$) have $z = -0.08$, -1.41 and 0.54, so the deviation does not persist across shot levels or degrees. We regard it as most likely a chance fluctuation but flag it for confirmation with more repetitions.

F. Structure across polynomial orders. Figure 9 resolves the error by order. P_0 has zero error at every degree, and for $n \geq 1$ the per-order RMSE at fixed N is nearly uniform (about 0.023–0.032 at $N = 1000$) because $\langle 1 - P_n^2 \rangle$ over the x grid lies between about 0.67 ($n = 1$) and 1 (large n). The observed values track the theoretical bars for $d = 12$, confirming that the model holds order by order and not only on average.

G. Input reconstruction and decoder efficiency. Decoded-input RMSE scales as $\sigma_d/\sqrt{N}$ with fitted prefactors $\sigma_d = 0.804, 0.451, 0.309, 0.227$ and 0.155 for $d = 2, 4, 6, 8, 12$. Going from degree 6 to 12 halves σ_d and therefore cuts the shots needed for a given accuracy by about a factor of four. The per-x error (Fig. 10, left) is not uniform: it is smallest near $x = \pm 1$, where features approach ± 1 and variances vanish, and largest in the interior.

Compared with the Cramér–Rao bound on interior points, the empirical decoded RMSE has a mean ratio of 0.999 (range 0.924–1.081; Fig. 10, right, and Table 5). Ratios slightly below 1 are possible because the decoder output is clipped to the valid range and the empirical estimate uses only five repetitions per x; ratios slightly above 1 reflect finite-sample scatter. No condition shows a systematic inefficiency: within sampling uncertainty the decoder extracts the information the measured features contain.

H. Shot budget for a target accuracy. Using the fitted prefactors, the number of shots required for a target decoded-input RMSE ε is $N \approx (\sigma_d/\varepsilon)^2$ (Table 6, Fig. 11). Values for $\varepsilon = 0.01$ lie within the measured range for $d \geq 6$ ($N \leq 5000$) and are validated by the data (e.g. $d = 6$ at $N = 1000$ gives 0.0102). Values outside the measured shot range are extrapolations of the $N^{-1/2}$ law and should be confirmed before use.

I. Worked example and API usage. The benchmark calls are reproduced by the following minimal snippet, which mirrors the script's use of the public API (Formulon 1.0.0):

```
from formulon.quantum_computing.legendre_encoder
import (
```

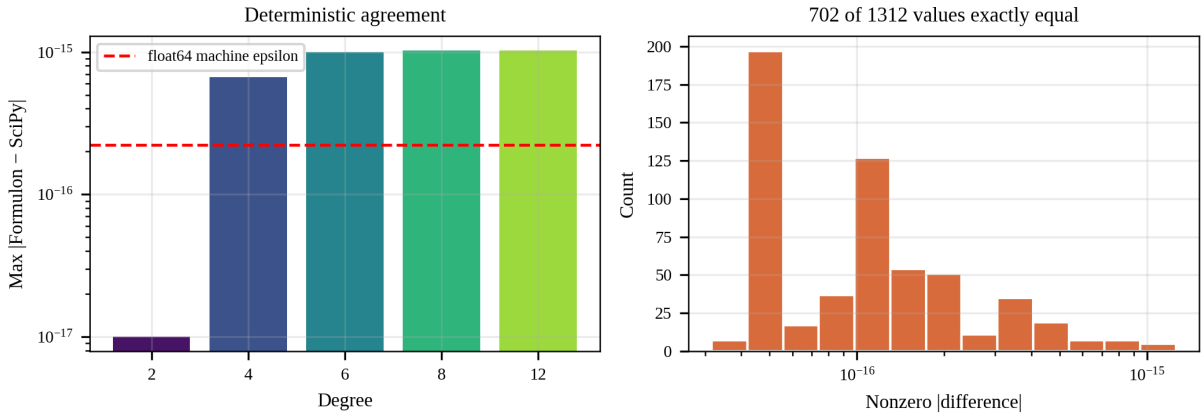


FIG. 2. Formulon-versus-SciPy agreement. Left: maximum absolute difference per degree (log scale) with float64 machine epsilon. Right: histogram of the nonzero differences.

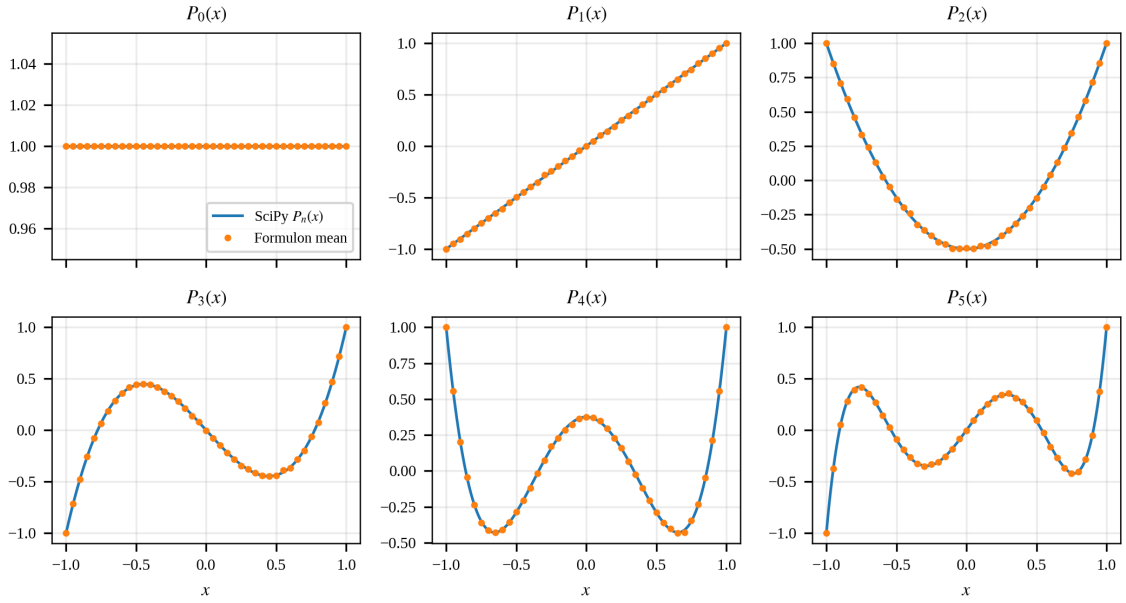


FIG. 3. Formulon measured mean (orange) versus SciPy $P_n(x)$ (blue) for degree 6 and 5000 shots, orders 0–5, averaged over five repetitions. Regenerated from `formulon_raw_results.csv` using the plotting logic of the benchmark script.

```
  legendre_features, measure_and_decode)
f = legendre_features(0.35, 8)   # deterministic
m = measure_and_decode(x=0.35, degree=8,
shots=500,
  xmin=-1.0, xmax=1.0, seed=7)
m.z_expectations              # measured features
m.decoded_x                 # reconstructed input
```

Sizing example. Suppose an application needs the input recovered to an RMSE of 0.01 with a degree-8 encoder. Table 6 gives $N \approx (\sigma_8/0.01)^2 \approx 516$ shots, with $\sigma_8 = 0.227$. The benchmark independently confirms this: at $d = 8$ and $N = 500$ the measured decoded-input RMSE is 0.0103, and at $N = 1000$ it is 0.0067, bracketing the target. For the same accuracy a degree-2 encoder would need about 6 500 shots, i.e. more than a factor of twelve more, which shows that the choice of degree matters more for the shot budget than the choice of any decoder refinement, given the decoder already saturates the Cramér–Rao bound.

Feature-level sizing. If instead the downstream model consumes the features and requires a per-feature RMSE below 0.02 at $d = 8$, Eq. (3) with the observed $\langle 1 - P_n^2 \rangle \approx 0.72$ gives $N \approx 0.72/0.02^2 \approx 1\,800$ shots; the measured RMSE at $N = 1000$ is 0.0271 and at $N = 5000$ is 0.0119, consistent with this estimate.

V. DISCUSSION

Correctness. The deterministic layer reproduces SciPy to round-off for all tested degrees up to 12, so the encoder's mathematics can be trusted as a reference feature map. The finite-shot layer then behaves like an ideal binomial sampler: the variance formula, the $N^{-1/2}$ scaling and the normality of the errors all hold quantitatively. This is strong evidence that the circuit constructed by Formulon encodes the features as intended and that the measurement-and-estimation path introduces no additional bias or excess variance on the noiseless

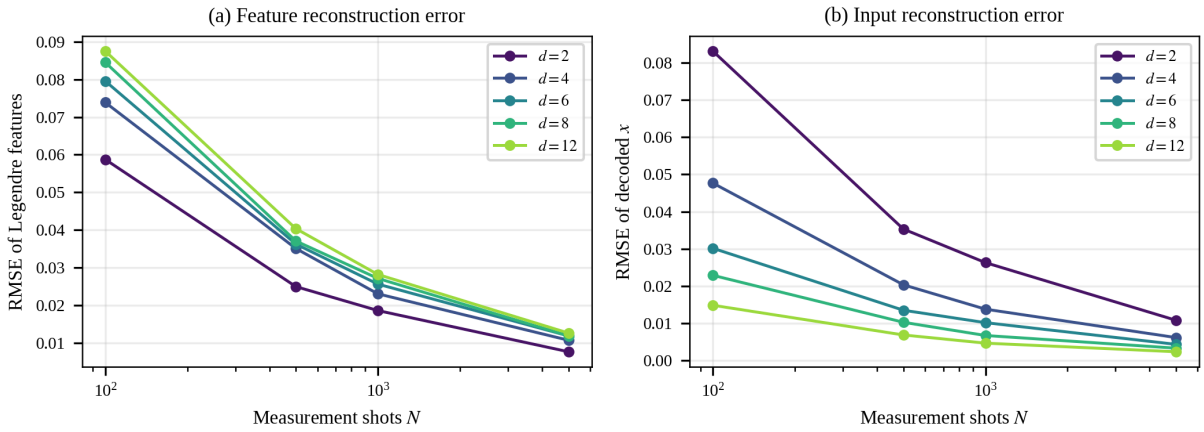


FIG. 4. Accuracy versus shot count for the five degrees: (a) feature RMSE and (b) decoded-input RMSE (data from the benchmark's summary CSV).

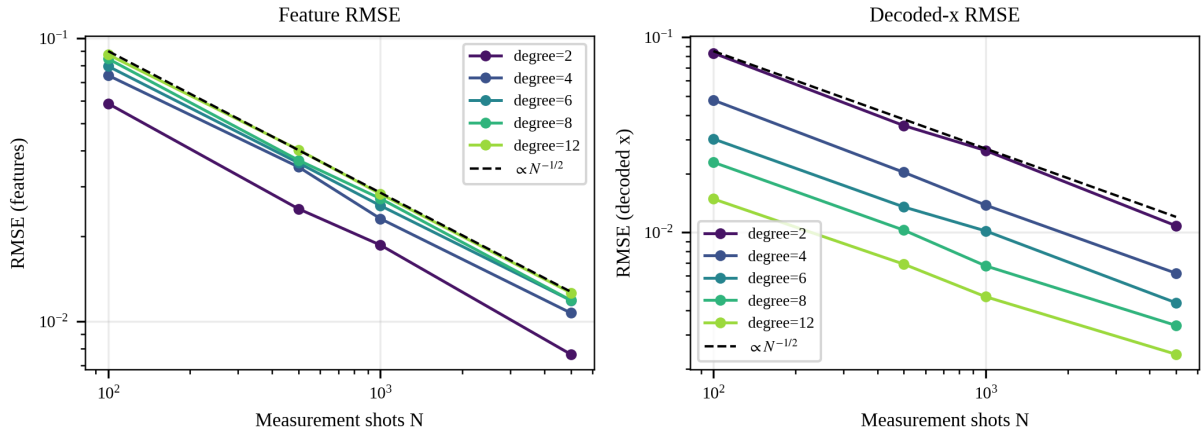


FIG. 5. Log–log view with an $N^{-1/2}$ reference (dashed). All curves are parallel to the reference, indicating the standard-quantum-limit scaling expected of shot noise.

simulator.

Degree trade-off. Practitioners often assume that raising the degree increases noise. The data show a more nuanced picture: per-feature noise increases mildly (feature RMSE $0.0588 \rightarrow 0.0875$ at $N = 100$) while decoded-input accuracy improves by a factor of about 4.5–5.6 between $d = 2$ and $d = 12$ (depending on N). If the downstream task consumes the features directly, the relevant penalty is the per-feature noise; if the task is reconstruction of x, higher degrees are beneficial at a fixed shot budget, with diminishing returns in absolute terms.

Decoder efficiency. Matching the Cramér–Rao bound within a few percent indicates that further decoder engineering would yield negligible statistical gain on this simulator; improvements must come from more shots, a different feature set, or noise-aware modelling on hardware. Conversely, the bound gives users a closed-form way to size shot budgets without running the simulation.

Practical recommendations. (1) Use Table 6 or $N \approx (\sigma_d/\varepsilon)^2$ to size shot budgets. (2) Expect the largest decoding errors at interior x and the smallest at the endpoints. (3) When comparing encoders, report both feature and input errors, since they move in opposite directions with degree. (4) Validate any hardware run against the ideal-simulator baseline given here, so excess error can be attributed to device noise rather than sampling.

VI. LIMITATIONS AND THREATS TO VALIDITY

- **Ideal simulator only.** PennyLane `default.qubit` with finite shots has no gate, readout or decoherence noise; hardware performance will differ.
- **Five repetitions per cell.** The script header recommends 30–100 repetitions for a final study. Cell-level estimates (especially the marginal bias at $d = 8$, $N = 500$ and the per-x curves) carry visible sampling uncertainty.
- **Seed construction.** The seed formula is unique within each shot level (1 025 distinct seeds per level) but not across levels: the 4 100 calls use only 3 485 distinct seed values because, for instance, $(d = 8, N = 500)$ and $(d = 12, N = 100)$ map to the same offset. Results at different N are therefore not strictly statistically independent. This does not affect within-cell statistics or the scaling conclusions, but future versions should hash all factors into

TABLE 3. Summary of all 20 (degree, shots) conditions from `summary_by_degree_and_shots.csv`. "Feat." denotes Legendre-feature error and "x" denotes decoded-input error. Feature records per cell are $205\,d$.

		Feature error				Decoded-x error		
d	N	MAE	RMSE	Max	Bias	MAE	RMSE	Max
2	100	0.0314	0.0588	0.2500	−0.0007	0.0627	0.0831	0.2500
2	500	0.0131	0.0250	0.1300	−0.0003	0.0263	0.0353	0.1300
2	1000	0.0094	0.0186	0.0840	+0.0011	0.0188	0.0263	0.0840
2	5000	0.0041	0.0077	0.0432	−0.0001	0.0082	0.0108	0.0430
4	100	0.0497	0.0739	0.2637	+0.0018	0.0336	0.0477	0.1600
4	500	0.0237	0.0351	0.1363	+0.0004	0.0152	0.0203	0.0620
4	1000	0.0156	0.0230	0.1045	−0.0002	0.0100	0.0138	0.0610
4	5000	0.0069	0.0107	0.0436	+0.0004	0.0042	0.0062	0.0240
6	100	0.0558	0.0796	0.3103	+0.0025	0.0216	0.0302	0.1070
6	500	0.0256	0.0363	0.1398	+0.0010	0.0100	0.0135	0.0490
6	1000	0.0180	0.0256	0.0997	−0.0010	0.0073	0.0102	0.0340
6	5000	0.0082	0.0119	0.0583	−0.0004	0.0031	0.0044	0.0180
8	100	0.0609	0.0845	0.3000	−0.0002	0.0168	0.0229	0.0830
8	500	0.0269	0.0371	0.1545	+0.0030	0.0077	0.0103	0.0410
8	1000	0.0197	0.0271	0.1093	−0.0009	0.0046	0.0067	0.0270
8	5000	0.0086	0.0119	0.0536	+0.0002	0.0024	0.0033	0.0100
12	100	0.0647	0.0875	0.4010	+0.0020	0.0110	0.0149	0.0560
12	500	0.0296	0.0404	0.1759	−0.0001	0.0052	0.0069	0.0230
12	1000	0.0211	0.0282	0.1009	+0.0000	0.0036	0.0047	0.0170
12	5000	0.0093	0.0126	0.0474	+0.0001	0.0017	0.0024	0.0100

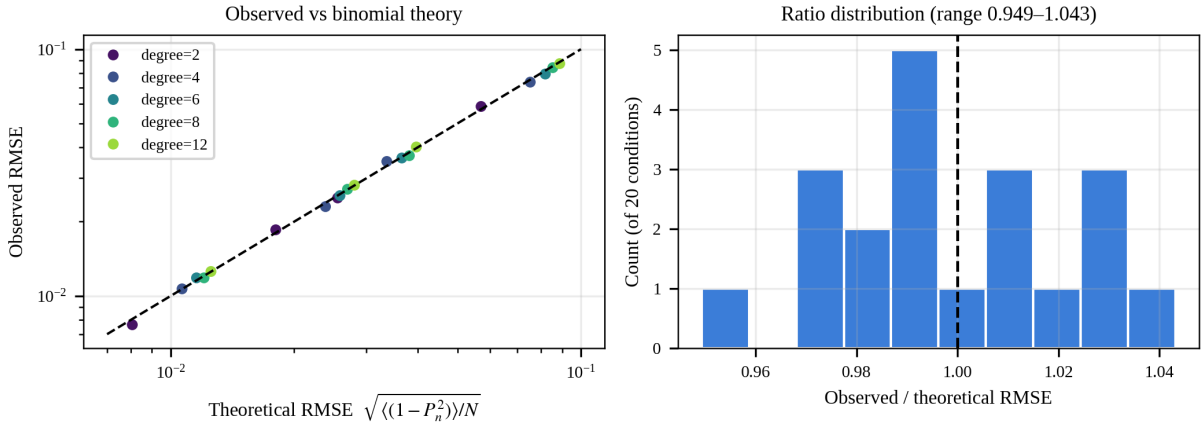


FIG. 6. Left: observed versus theoretical feature RMSE for the 20 conditions. Right: distribution of the observed/theoretical ratio.

TABLE 4. Ratio of observed to theoretical feature RMSE (1 = perfect agreement with binomial shot noise).

d	N=100	N=500	N=1000	N=5000
2	1.031	0.979	1.032	0.949
4	0.982	1.043	0.969	1.007
6	0.974	0.992	0.992	1.026
8	0.993	0.974	1.008	0.987
12	0.987	1.018	1.005	1.006

the seed.

- **Uniform x grid.** A different input distribution changes the averages $\langle 1 - P_n^2 \rangle$ and hence absolute RMSE values, though not the $N^{-1/2}$ law.
- **Limited range.** Intermediate or very large degrees and shot counts above 5000 were not tested; budgets outside the range are extrapolations.
- **Black-box decoder.** The Cramér–Rao comparison uses only measured error statistics and the analytic Fisher information. It assumes independent, approximately Gaussian feature errors, supported by Sec. IV E but only approximate

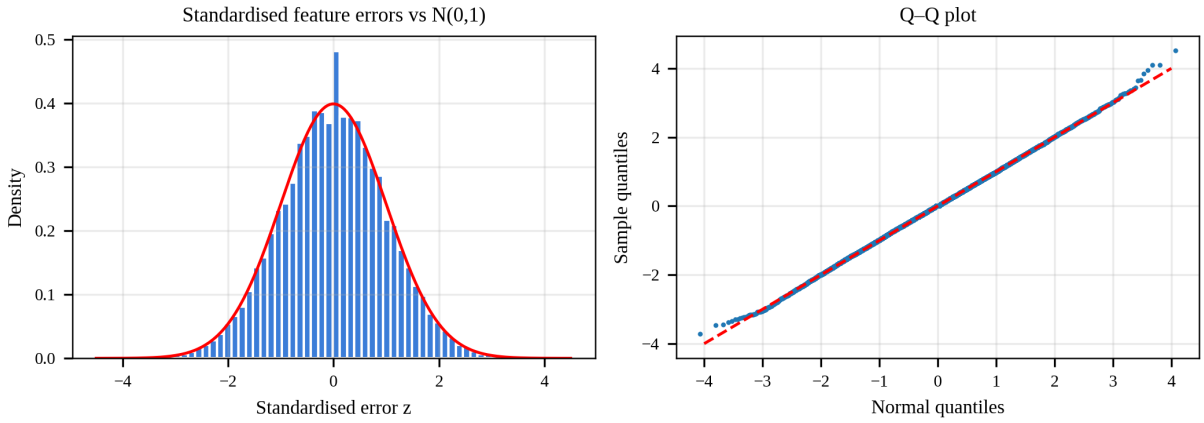


FIG. 7. Standardised feature errors compared with the standard normal density (left) and Q–Q plot (right).

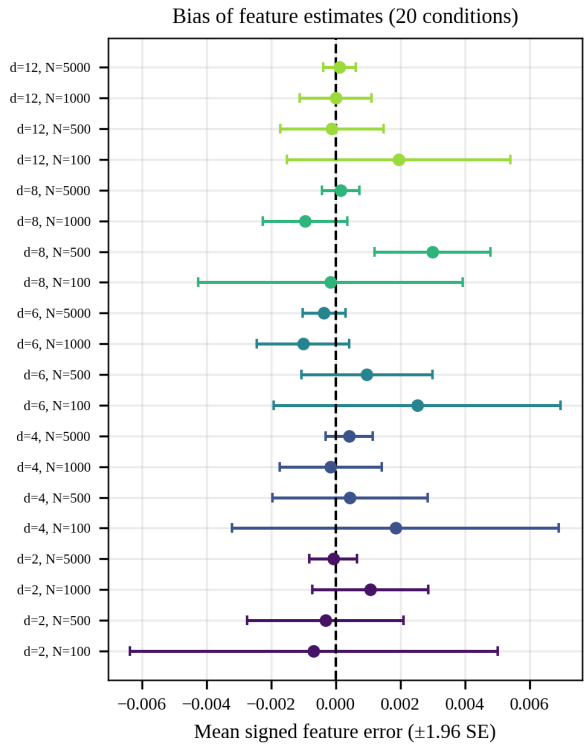


FIG. 8. Mean signed feature error with 95 % confidence intervals for all 20 conditions. Intervals overlap zero except one marginal case ($d = 8$, $N = 500$).

at $N = 100$.

- **Single software stack and date.** One Colab session with one set of library versions was used.
- **No quantum-advantage claim.** The benchmark checks numerical correctness and statistical efficiency only.

VII. REPRODUCIBILITY, DATA AND CODE AVAILABILITY

The study can be re-run in a fresh Google Colab notebook:

```
!pip -q install "formulon-physics==1.0.0"
pennylane scipy pandas numpy matplotlib
%run formulon_legendre_benchmark_colab.py
```

The script writes `formulon_legendre_benchmark_results/` containing the raw records (`formulon_raw_results.csv`, 26 240 rows), the 20-row summary (`summary_by_degree_and_shots.csv`), the deterministic comparison (`deterministic_formulon_vs_scipy.csv`, 1,312 rows), `benchmark_metadata.json`, and the figures; all accompany this preprint on Zenodo (doi:10.5281/zenodo.23753750). The additional analyses reported here (shot-noise ratios, z-score diagnostics, bias intervals, Cramér–Rao comparison, shot budget) were

TABLE 5. Decoder efficiency on interior points ($|x| < 0.95$) for $N = 100, 1000, 5000$; all 20 conditions appear in Fig. 10.

d	N	Empirical RMSE	CRLB RMSE	Ratio
2	100	0.08718	0.08456	1.031
2	1000	0.02763	0.02674	1.033
2	5000	0.01135	0.01196	0.949
4	100	0.05011	0.04729	1.060
4	1000	0.01451	0.01495	0.970
4	5000	0.00650	0.00669	0.972
6	100	0.03174	0.03206	0.990
6	1000	0.01072	0.01014	1.057
6	5000	0.00459	0.00453	1.012
8	100	0.02409	0.02428	0.992
8	1000	0.00709	0.00768	0.924
8	5000	0.00351	0.00343	1.022
12	100	0.01563	0.01634	0.956
12	1000	0.00492	0.00517	0.953
12	5000	0.00250	0.00231	1.081

TABLE 6. Estimated shots required for a target decoded-input RMSE ε, $N \approx (\sigma_d/\varepsilon)^2$.

d	σ_d	ε=0.05	0.02	0.01	0.005
2	0.804	259	1,618	6,470	25,881
4	0.451	81	509	2,037	8,149
6	0.309	38	238	953	3,810
8	0.227	21	129	516	2,063
12	0.155	10	60	239	957

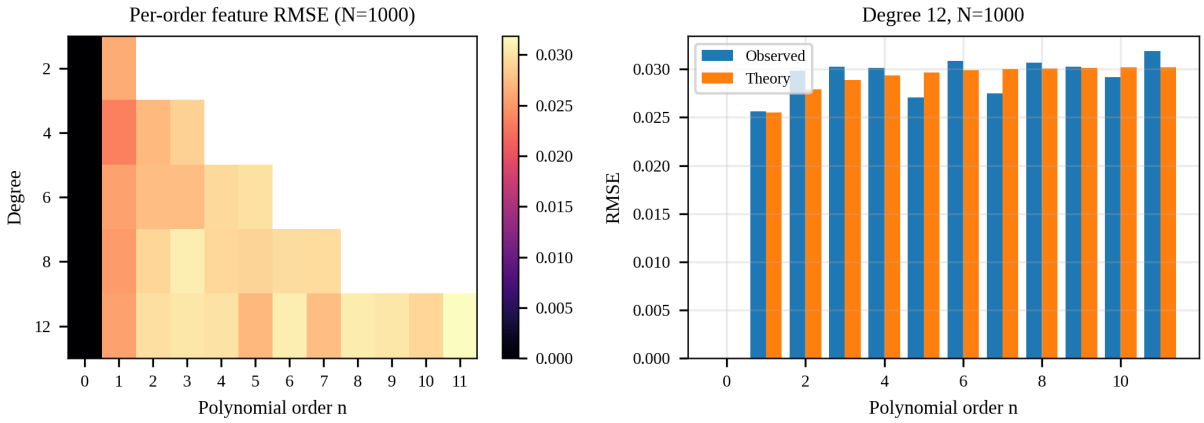


FIG. 9. Left: per-order feature RMSE at $N = 1000$ for each degree. Right: observed versus theoretical RMSE per order for $d = 12$ at $N = 1000$.

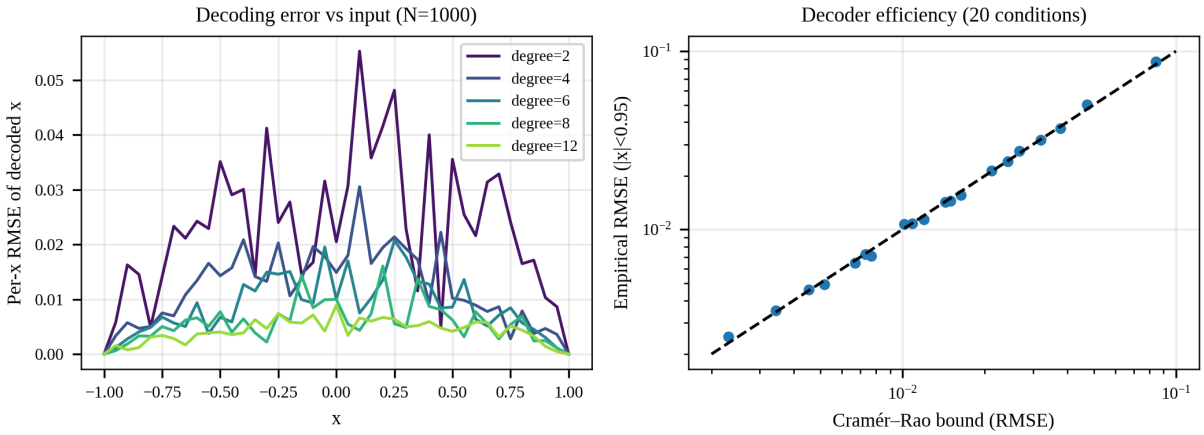


FIG. 10. Left: per-x RMSE of the decoded input at $N = 1000$. Right: empirical decoded RMSE ($|x| < 0.95$) versus the Cramér–Rao bound for all 20 conditions; the dashed line is equality.

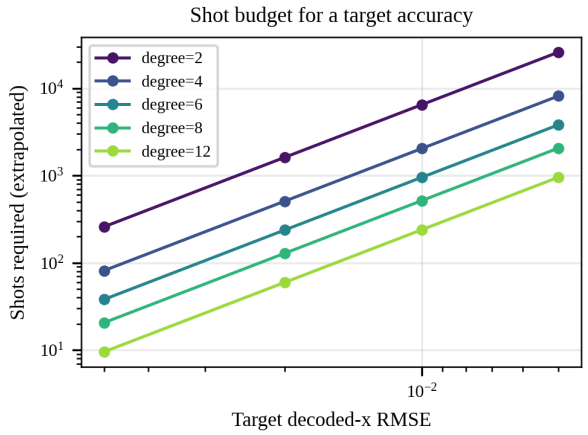


FIG. 11. Shot budget versus target accuracy for each degree (extrapolated from the fitted $N^{-1/2}$ prefactors).

computed from the released CSV files only and require no re-simulation. With fixed seeds, re-running with identical library versions should reproduce the CSV files; with different versions small differences in sampled values are possible although the statistical conclusions should hold.

AI-assistance disclosure. Generative AI (OpenAI ChatGPT) assisted in drafting and documenting the benchmark script, as stated in its header and metadata. Anthropic's Claude assisted in the statistical analysis of the released data, figure generation and typesetting of this report. The author is responsible for verification, execution of the Colab experiments, interpretation and all conclusions.

Competing interests. The author is the developer of the Formulon Physics library under test; the benchmark is therefore a self-evaluation and independent replication is encouraged.

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VIII. CONCLUSION

On a noise-free PennyLane simulator, Formulon Physics 1.0.0's Legendre encoder reproduces SciPy's Legendre values to 1.03×10^{-15}, and its finite-shot measurements behave as ideal binomial estimators: feature RMSE matches $\sqrt{\langle 1 - P_n^2 \rangle / N}$ to about $\pm 5\,\%$, scales as $N^{-1/2}$, is normally distributed and is unbiased within the resolution of five repetitions. The input decoder attains the Cramér–Rao bound within roughly $\pm 8\,\%$ and improves strongly with degree (σ_d from 0.804 at $d = 2$ to 0.155 at $d = 12$). These results pro-

vide a statistical baseline against which future hardware or noise-model studies of the encoder can be compared. Future work should increase the repetition count, repair the seed construction, add a Qiskit cross-check, and test noisy-simulator and hardware backends.

REFERENCES

[1] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions* (National Bureau of Standards, Washington, DC, 1964), Chs. 8 and 22.

[2] V. Bergholm *et al.*, PennyLane: Automatic differentiation of hybrid quantum-classical computations, arXiv:1811.04968 (2018).

[3] H. Cramér, *Mathematical Methods of Statistics* (Princeton University Press, Princeton, 1946).

[4] C. R. Harris *et al.*, Array programming with NumPy, Nature **585**, 357 (2020).

[5] J. D. Hunter, Matplotlib: A 2D graphics environment, Comput. Sci. Eng. **9**, 90 (2007).

[6] W. McKinney, Data structures for statistical computing in Python, in *Proc. 9th Python in Science Conf.* (2010), p. 56.

[7] C. R. Rao, Information and the accuracy attainable in the estimation of statistical parameters, Bull. Calcutta Math. Soc. **37**, 81 (1945).

[8] P. Virtanen *et al.*, SciPy 1.0: fundamental algorithms for scientific computing in Python, Nat. Methods **17**, 261 (2020).

[9] Formulon Physics, version 1.0.0, Python package (PyPI: `formulon-physics`).

[10] S. Kaleeswarar S, Benchmark script, raw data and metadata for the Formulon Legendre encoder (Google Colab, 29 October 2026), Zenodo, doi:10.5281/zenodo.23753750.

APPENDIX A: RECORDED ENVIRONMENT METADATA

TABLE 7. Environment and configuration recorded in `benchmark_metadata.json`.

Field	Value
`formulon_version`	1.0.0
`python_version`	3.13.18 (Colab)
`platform`	Linux-6.6.159+-x86_64-with-glibc2.39
`numpy_version`	2.1.3
`pandas_version`	2.2.3
`scipy_version`	1.16.3
`pennylane_version`	0.45.1
`backend`	PennyLane default.qubit
`x_points`	41
`degrees`	2, 4, 6, 8, 12
`shot_levels`	100, 500, 1000, 5000
`repetitions`	5
`seed_base`	20261028

APPENDIX B: PER-ORDER FEATURE RMSE AT $N = 5000$

APPENDIX C: ZENODO RECORD METADATA

TABLE 9. Metadata of the associated Zenodo record.

Field	Entry
Resource type	Publication → Preprint
DOI	10.5281/zenodo.23753750
Creator	Sri Kaleeswarar S, ORCID 0009-0005-6360-2345
Affiliations	Scott Christian College, Nagercoil (Reg. No. 250075); Open Science Labs; Formulon Physics
Date	29 October 2026
Keywords	Formulon Physics; Legendre polynomials; shot noise; PennyLane; reproducibility
Licence	CC BY 4.0 (suggested)
Versions	formulon-physics 1.0.0; PennyLane 0.45.1

APPENDIX D: DERIVATIONS USED IN THE TEXT

Binomial variance. Let $Y_i \in \{+1,-1\}$ with $\Pr(Y_i=+1) = (1+p)/2$. Then $\mathbb{E}[Y_i] = p$ and $\mathbb{E}[Y_i^2] = 1$, so $\mathrm{Var}(Y_i) = 1-p^2$. The sample mean $\hat{p} = N^{-1}\sum_i Y_i$ over independent shots is unbiased with $\mathrm{Var}(\hat{p}) = (1-p^2)/N$, which is Eq. (2). The error vanishes identically for $p = \pm 1$.

Grid average of P_n^2. By orthogonality, $\int_{-1}^{1} P_n(x)^2\,dx = 2/(2n+1)$, so for a uniform input distribution on $[-1,1]$ one has $\langle P_n^2 \rangle = 1/(2n+1)$ and $\langle 1-P_n^2 \rangle = 2n/(2n+1)$. This equals $2/3$ for $n=1$, $4/5$ for $n=2$ and tends to 1 for large n; the 41-point grid reproduces these values to within a few per cent. Substituting into Eq. (3) gives per-order RMSE values between $\sqrt{2/3N}$ and $\sqrt{1/N}$, which at $N = 5000$ is 0.0115–0.0141, matching Table 8.

Fisher information. For independent features with $\hat{f}_n \sim \mathcal{N}(P_n(x), (1-P_n^2)/N)$, the log-likelihood derivative with respect to x is $\sum_n [\hat{f}_n - P_n]\,P_n'/\sigma_n^2$ plus a variance-derivative term that is $O(1/N)$ smaller. The leading Fisher information is $\sum_n (P_n')^2/\sigma_n^2 = N\sum_n (P_n')^2/(1-P_n^2)$, Eq. (4).

Degree-2 identity. For $d = 2$ the feature vector is $(1, \hat{P}_1)$ with $P_1 = x$; the decoder output equals $\hat{P}_1$ (up to clipping), so $\mathrm{RMSE}_x = \mathrm{RMSE}(\hat{P}_1)$. Averaging over the two features gives $\mathrm{RMSE}_{\mathrm{feat}} = \sqrt{(0+\mathrm{RMSE}_x^2)/2} = \mathrm{RMSE}_x/\sqrt{2}$. The data confirm this: $0.08309/\sqrt{2} = 0.05875$ at $N = 100$, matching the tabulated 0.05875.

APPENDIX E: SCALING FITS AND SEED STATISTICS

TABLE 10. Least-squares log–log slopes of RMSE versus N (theory: -0.5) and fitted decoded-input prefactor σ_d (mean of RMSE$\cdot\sqrt{N}$ over the four shot levels).

d	Feature slope	Decoded-x slope	σ_d
2	−0.518	−0.518	0.804
4	−0.497	−0.523	0.451
6	−0.487	−0.492	0.309
8	−0.500	−0.496	0.227
12	−0.496	−0.471	0.155

Across the 4 100 calls the script produced 3,485 distinct seed values; within every shot level all 1,025 seeds are distinct, whereas 1,230 calls share a seed value with at least one call at a different shot level. Because the shot count differs be-

TABLE 8. Per-order feature RMSE at 5000 shots. Order 0 is exactly zero; the remaining orders lie in the range 0.0108–0.0147, close to the theoretical $\sqrt{0.67–1}/\sqrt{5000} = 0.0116–0.0141$.

d	P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	P_{11}
2	0.0000	0.0108										
4	0.0000	0.0123	0.0121	0.0127								
6	0.0000	0.0125	0.0134	0.0125	0.0133	0.0132						
8	0.0000	0.0109	0.0116	0.0125	0.0124	0.0142	0.0137	0.0133				
12	0.0000	0.0109	0.0135	0.0135	0.0133	0.0137	0.0139	0.0124	0.0147	0.0124	0.0129	0.0135

tween such calls, the sampled outcome streams are generated for different circuit executions, but the underlying pseudo-random stream is the same, so correlation between cells at different N cannot be excluded. A collision-free alternative is to derive the seed from a hash of (i_x, d, N, r).

APPENDIX F: BIAS TESTS FOR ALL 20 CONDITIONS

TABLE 11. Mean signed feature error, standard error and z-statistic per condition. Under the null hypothesis of no bias, about one of 20 values is expected to exceed $|z| = 2$; the largest observed is $|z| = 3.28$ ($d = 8, N = 500$), with Bonferroni-adjusted $p \approx 0.02$.

d	N	Records	Mean error	SE	z
2	100	410	−0.00068	0.00290	−0.24
2	500	410	−0.00032	0.00123	−0.26
2	1000	410	+0.00107	0.00092	+1.17
2	5000	410	−0.00008	0.00038	−0.21
4	100	820	+0.00185	0.00258	+0.72
4	500	820	+0.00044	0.00123	+0.36
4	1000	820	−0.00016	0.00081	−0.20
4	5000	820	+0.00041	0.00037	+1.11
6	100	1230	+0.00252	0.00227	+1.11
6	500	1230	+0.00096	0.00103	+0.93
6	1000	1230	−0.00101	0.00073	−1.38
6	5000	1230	−0.00036	0.00034	−1.08
8	100	1640	−0.00017	0.00209	−0.08
8	500	1640	+0.00300	0.00091	+3.28
8	1000	1640	−0.00094	0.00067	−1.41
8	5000	1640	+0.00016	0.00029	+0.54
12	100	2460	+0.00196	0.00176	+1.11
12	500	2460	−0.00012	0.00081	−0.15
12	1000	2460	+0.00000	0.00057	+0.00
12	5000	2460	+0.00012	0.00025	+0.46

APPENDIX G: DECODER EFFICIENCY FOR ALL 20 CONDITIONS

TABLE 12. Empirical decoded-input RMSE on interior points ($|x| < 0.95$) versus the Cramér–Rao bound for every condition.

d	N	Empirical RMSE	CRLB RMSE	Ratio
2	100	0.08718	0.08456	1.031
2	500	0.03706	0.03782	0.980
2	1000	0.02763	0.02674	1.033
2	5000	0.01135	0.01196	0.949
4	100	0.05011	0.04729	1.060
4	500	0.02138	0.02115	1.011
4	1000	0.01451	0.01495	0.970
4	5000	0.00650	0.00669	0.972
6	100	0.03174	0.03206	0.990
6	500	0.01422	0.01434	0.991
6	1000	0.01072	0.01014	1.057
6	5000	0.00459	0.00453	1.012
8	100	0.02409	0.02428	0.992
8	500	0.01082	0.01086	0.997
8	1000	0.00709	0.00768	0.924
8	5000	0.00351	0.00343	1.022
12	100	0.01563	0.01634	0.956
12	500	0.00725	0.00731	0.991
12	1000	0.00492	0.00517	0.953
12	5000	0.00250	0.00231	1.081